

Native Fedora · Ryzen 7800X3D · RTX 4090 24 GB · full bounded roadmap · M00-M23 + C0-C4

fedora-roadmap-20261008T022526329292Z | 10 completed stages | 3 paired training seeds

experiment records

74,149 science + 1 report inventory

individual check records

Every occurrence is mapped below

software tests passed

1 skip - 0 failures/errors

independent confirmation fields

12 continuum fields; 6 policy fields

Scores reflect required check weight met, not model quality or proof.
One mechanism can link to many repeated experiment records.

ID	DECLARED MECHANISM	SCORE	KNOWN
M00	Headroom map and fused classical reference	61	97%
M01	Consistent spatial target and two-resolution dis...	68	82%
M02	Exact finite-step quadratic DF defect	61	92%
M03	Physical factors plus learned rank-1/rank-2 rema...	60	92%
M04	Signed nonlinear sources before output compression	60	92%
M05	Product-consistent anti-aliasing and grid-scaled...	82	94%
M06	Stiffness-aware trust envelope for the correction	59	91%
M07	Conditional solver bank with a classical safe path	68	81%
M08	Cheap embedded estimator and stability-weighted ...	68	81%
M09	Function-level calibrated total-error and select...	68	81%
M10	Unequal-step generator consistency and intermed...	61	91%
M11	DF-tuned defect loss with regime and maximum-err...	60	92%
M12	Dynamic feasible mean-variance closure	93	93%
M13	Dimensionally consistent multi-band/fractional f...	59	92%
M14	Geometry-aware and anisotropic transport	82	94%
M15	Cost-aware algebraic reuse, fusion and batching	67	81%
M16	Faithful neural comparison and stronger classical...	44	81%
M17	Factor-isolated stress and coefficient-transfer ...	84	94%
M18	Fresh independent fields and hierarchical infer...	69	83%
M19	PDE and boundary portability ladder	82	94%
M20	Order-constrained learned embedded stages	65	92%
M21	Physical monotonicity and growth-envelope diagno...	81	92%
M22	Effective configuration, provenance and reproduc...	93	93%
M23	Dedicated cubic Volterra remainder	87	89%
C0	Prerequisite screen	74	74%
C1	Primary compact finite-step residual	61	92%
C2	Operational error/cost-controlled hybrid	62	90%
C3	Finite-amplitude side branch	94	95%
C4	Later transfer/geometry branch	82	99%

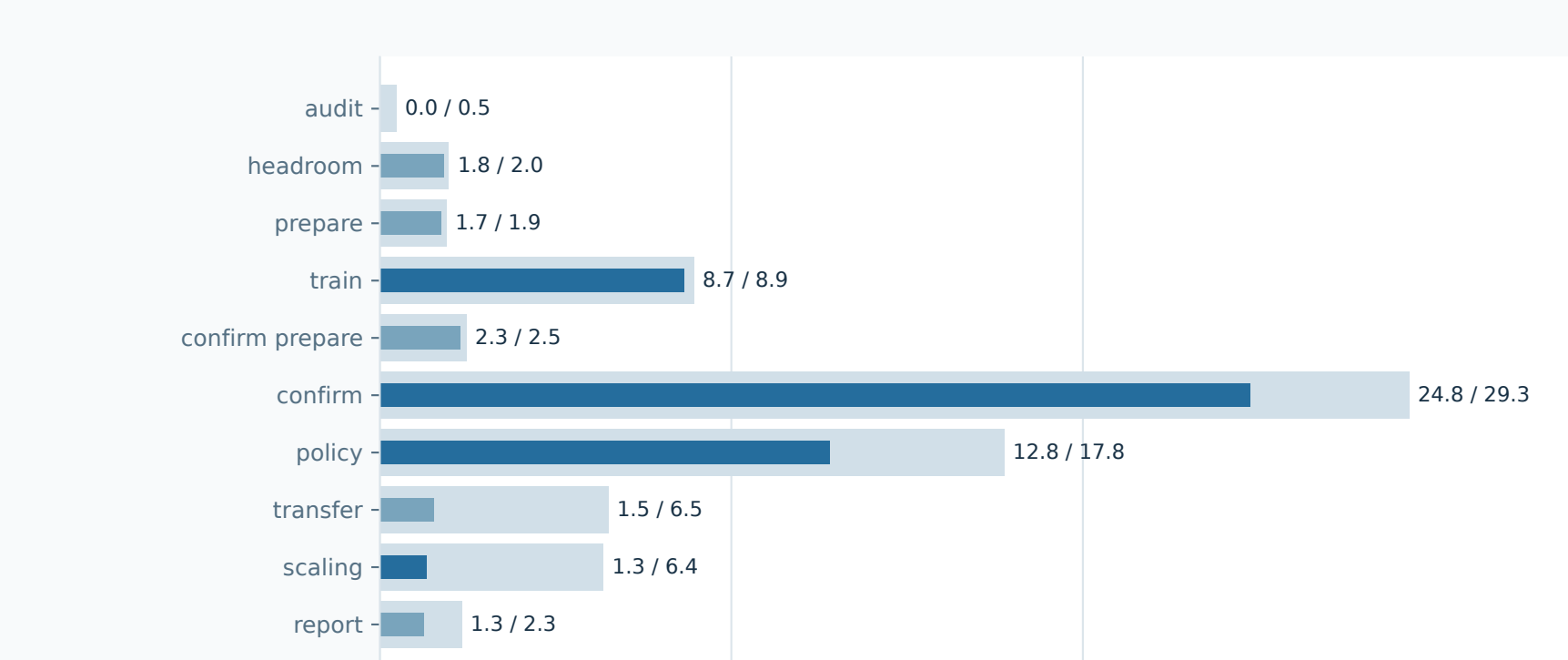
Recorded aggregate verdicts: 26 BAD, 3 NA. Names and exact counts: companion CSV.

Cells = fraction meeting BOTH RMS and maximum error targets.
Discrete and continuum targets differ; repeated endpoints are paired.

	2e-4	2e-5	2e-6	2e-4	2e-5	2e-6
source	97	53	28	70	26	8
rank1	100	75	37	71	29	8
rank2	100	75	37	71	28	8
source_postcompression	98	51	29	70	27	7
source_trust	96	51	28	70	27	8
source_consistency	97	53	28	70	26	8
source_df_loss	96	52	28	70	26	8
source_df_loss_shared	96	52	28	70	26	8
source_multiband	96	52	28	69	26	8
cheap_fno	98	53	28	72	26	8
deep_fno	96	53	28	70	27	8
direct_fno	0	0	0	0	0	0
c1_rank0	99	90	57	67	27	8
c1_rank1	99	91	58	67	26	8
c1_rank2	99	91	58	68	26	8
c2_rank2	98	84	58	68	27	9
source_embedded	100	100	93	66	28	7
etdrk4	62	54	27	33	11	0
df_quad2	100	92	58	68	27	8
df_base	98	53	28	72	26	8
rf_base	88	44	24	55	21	2
gl3_fused	100	88	70	66	28	8

¹ Median discrete latency across measured schedules (different work per schedule); descriptive, not an accuracy-matched ranking.

Times show science vs. complete Slurm allocation (excludes queue).
Policy costs include rejected attempts and fallback.



Science 56.2 min / allocated 78.0 min / GPU allocated 62.4 min

DEPLOYED POLICY · 984 endpoints · 6 independent fields

900 jointly accurate / 84 false accepts
412 accurate fallbacks / no formal coverage guarantee
12.99x aggregate classical cost; accuracy is not matched

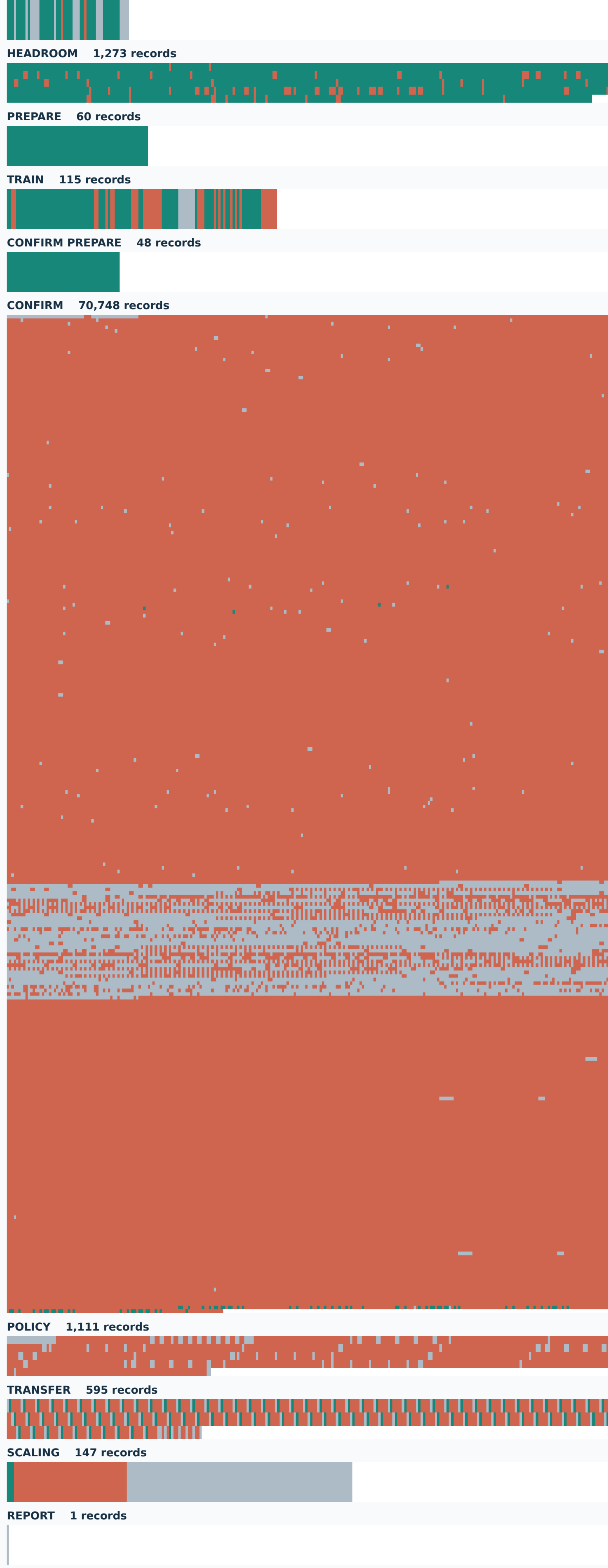
CHECK PURPOSE	GOOD	BAD	NA
correctness	90.173	0	0
math	62.258	4.706	10.566
gap	204.209	116.686	334
utility	14.258	73.969	42.538

GOOD = declared checks met • BAD = at least one required check failed • NA = evidence incomplete • scores are evidence attainment, not probabilities.

The original accuracy-matched frontiers mixed horizons in 1,412 / 22,032 rows. Raw scores below are preserved; the companion PDF adds corrected comparisons.

Left-to-right, then next row. 256 columns per stage. White = unused padding.

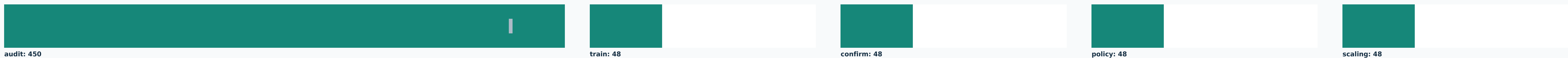
AUDIT 52 records



GOOD BAD NA

ALL EXPERIMENTS GOOD 1,573 / BAD 66,173 / NA 6,404

641 passed, 1 optional Tower-checkout skip; four repetitions of 48 GPU tests remain separate occurrences.



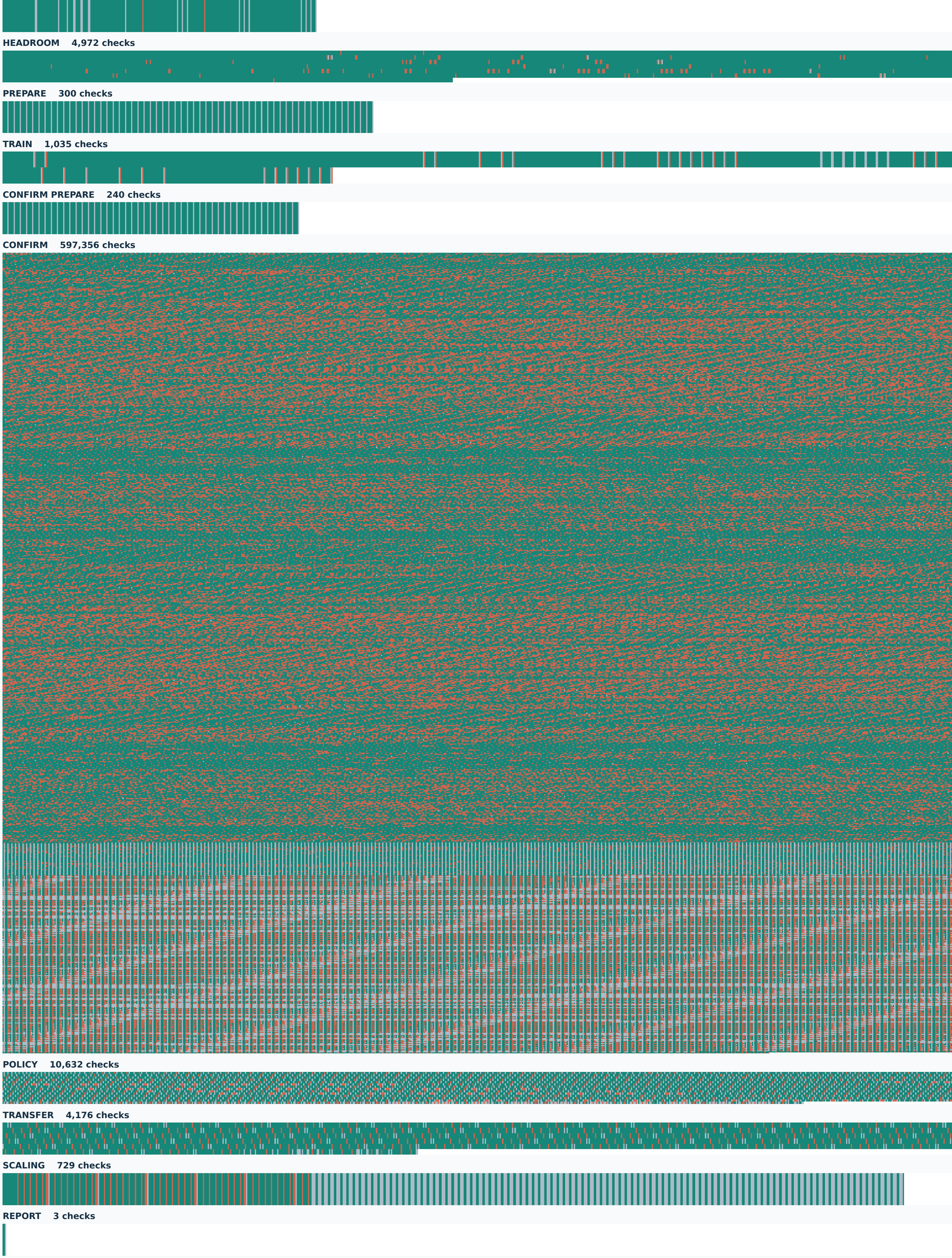
TRACEABILITY Each cell maps to its exact experiment/check/test identity, score, operands and source in the companion compressed CSV indexes.

Stage-local coordinates are zero-based. Colored cells are observations, not independent samples. A numerical pass is not a mathematical proof; these internal FNO controls are not an official-paper reproduction.

Source: integrity-checked uploaded full-run archive. Read-only visualization; no models retrained, no source observations modified. See atlas-manifest.json for source hashes and coverage audit.

Left-to-right, then next row. 768 columns per stage. No sampling or aggregation.

AUDIT 254 checks



ALL CHECKS GOOD 370,898 / BAD 195,361 / NA 53,438

BAD can reflect cost alone: 394 transfer records fail utility checks, not trajectory correctness.

HORIZON-MATCHED COMPARISONS / 0.0002

Offline reanalysis of existing timings: same target track, same final time, same parent / grid / paired seed.

Median comparator latency ÷ candidate latency. Above 1 = candidate faster. Post-hoc feasible schedule selection; excludes offline training cost.

DISCRETE · T=0.27				DISCRETE · T=0.81			CONTINUUM · T=0.27			CONTINUUM · T=0.81			
source_postcompression	source	1.02× n=288	1.90× n=288	0.32× n=288	1.03× n=288	1.91× n=288	0.29× n=288	1.02× n=102	1.90× n=102	0.32× n=102	1.02× n=114	1.91× n=114	0.29× n=114
	rank1	0.55× n=288	1.02× n=288	0.17× n=288	0.55× n=288	1.02× n=288	0.16× n=288	0.55× n=102	1.02× n=102	0.17× n=102	0.55× n=114	1.02× n=114	0.16× n=114
	rank2	0.41× n=288	0.76× n=288	0.13× n=288	0.40× n=288	0.75× n=288	0.12× n=288	0.41× n=102	0.76× n=102	0.13× n=102	0.40× n=114	0.76× n=114	0.12× n=114
	source_trust	0.92× n=288	1.70× n=288	0.29× n=288	0.92× n=288	1.71× n=288	0.26× n=288	0.92× n=102	1.70× n=102	0.29× n=102	0.92× n=114	1.71× n=114	0.26× n=114
	source_consistency	0.80× n=288	1.49× n=288	0.25× n=288	0.80× n=288	1.50× n=288	0.23× n=288	0.80× n=102	1.49× n=102	0.25× n=102	0.80× n=114	1.50× n=114	0.23× n=114
	source_df_loss	1.02× n=288	1.90× n=288	0.32× n=288	1.03× n=288	1.91× n=288	0.29× n=288	1.02× n=102	1.90× n=102	0.32× n=102	1.02× n=114	1.92× n=114	0.29× n=114
	source_df_loss_shared	1.02× n=288	1.90× n=288	0.32× n=288	1.03× n=288	1.91× n=288	0.29× n=288	1.02× n=102	1.90× n=102	0.32× n=102	1.02× n=114	1.91× n=114	0.29× n=114
	source_multiband	0.74× n=288	1.38× n=288	0.23× n=288	0.74× n=288	1.38× n=288	0.21× n=288	0.74× n=102	1.38× n=102	0.23× n=102	0.74× n=114	1.38× n=114	0.21× n=114
	cheap_fno	1.00× n=288	1.85× n=288	0.32× n=288	1.00× n=288	1.87× n=288	0.29× n=288	1.00× n=102	1.86× n=102	0.32× n=102	1.00× n=114	1.87× n=114	0.29× n=114
	deep_fno	0.54× n=288	1.00× n=288	0.17× n=288	0.54× n=288	1.00× n=288	0.15× n=288	0.54× n=102	1.00× n=102	0.17× n=102	0.54× n=114	1.00× n=114	0.15× n=114
	direct_fno	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
	c1_rank0	0.50× n=288	0.92× n=288	0.16× n=288	0.49× n=288	0.92× n=288	0.14× n=288	0.50× n=102	0.92× n=102	0.16× n=102	0.49× n=114	0.92× n=114	0.14× n=114
source_embedded	c1_rank1	0.30× n=288	0.55× n=288	0.09× n=288	0.29× n=288	0.55× n=288	0.08× n=288	0.30× n=100	0.55× n=100	0.09× n=100	0.29× n=114	0.55× n=114	0.08× n=114
	c1_rank2	0.25× n=288	0.46× n=288	0.08× n=288	0.25× n=288	0.46× n=288	0.07× n=288	0.25× n=100	0.46× n=100	0.08× n=100	0.25× n=114	0.46× n=114	0.07× n=114
	c2_rank2	0.23× n=288	0.43× n=288	0.07× n=288	0.23× n=288	0.43× n=288	0.07× n=288	0.23× n=102	0.43× n=102	0.07× n=102	0.23× n=114	0.43× n=114	0.07× n=114
	source_embedded	0.19× n=288	0.36× n=288	0.06× n=288	0.18× n=288	0.35× n=288	0.06× n=288	0.12× n=90	0.23× n=90	0.04× n=90	0.12× n=114	0.23× n=114	0.04× n=114
cheap fno				cheap fno			cheap fno			cheap fno			
deep fno				deep fno			deep fno			deep fno			
best classical				best classical			best classical			best classical			

Scope and limits

Both RMS and maximum error must be ≤ 0.0002. n is eligible paired measurement cells, not independent fields. Discrete: 24 independent fields; continuum: 12.

The 3 paired training seeds, 2 grids and 2 coefficient pairs reuse fields. Timing differences of only a few percent are not demonstrated robust speedups.

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Policy: 84 continuum false accepts remain failures. The reported formal-policy break-even timing uses identical fallback work and is confounded by first-call timing; it is not shown as a win.

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HORIZON-MATCHED COMPARISONS / 2e-05

Offline reanalysis of existing timings: same target track, same final time, same parent / grid / paired seed.

Median comparator latency ÷ candidate latency. Above 1 = candidate faster. Post-hoc feasible schedule selection; excludes offline training cost.

DISCRETE · T=0.27			DISCRETE · T=0.81			CONTINUUM · T=0.27			CONTINUUM · T=0.81				
source_postcompression	source	1.03× n=192	1.90× n=192	0.32× n=192	1.02× n=252	1.91× n=252	0.29× n=252	1.02× n=32	1.90× n=32	0.32× n=30	1.02× n=90	1.91× n=90	0.29× n=90
	rank1	0.56× n=192	1.03× n=192	0.22× n=288	0.55× n=252	1.02× n=252	0.16× n=288	0.55× n=33	1.02× n=36	0.18× n=42	0.55× n=84	1.02× n=84	0.16× n=84
	rank2	0.41× n=192	0.76× n=192	0.16× n=288	0.40× n=252	0.75× n=252	0.12× n=288	0.41× n=33	0.75× n=36	0.13× n=42	0.40× n=84	0.76× n=84	0.12× n=84
	source_trust	0.92× n=192	1.70× n=192	0.29× n=192	0.92× n=252	1.71× n=252	0.26× n=252	0.92× n=33	1.70× n=34	0.29× n=30	0.92× n=90	1.71× n=90	0.26× n=90
	source_consistency	0.80× n=192	1.49× n=192	0.25× n=192	0.80× n=252	1.50× n=252	0.23× n=252	0.80× n=32	1.49× n=32	0.26× n=30	0.80× n=90	1.50× n=90	0.23× n=90
	source_df_loss	1.02× n=192	1.90× n=192	0.32× n=192	1.03× n=252	1.91× n=252	0.29× n=252	1.02× n=32	1.90× n=32	0.33× n=30	1.03× n=90	1.91× n=90	0.29× n=90
	source_df_loss_shared	1.02× n=192	1.90× n=192	0.32× n=192	1.02× n=252	1.91× n=252	0.29× n=252	1.02× n=32	1.89× n=32	0.32× n=30	1.02× n=90	1.91× n=90	0.29× n=90
	source_multiband	1.02× n=192	1.90× n=192	0.32× n=192	1.02× n=252	1.91× n=252	0.29× n=252	1.03× n=32	1.90× n=32	0.33× n=30	1.02× n=90	1.91× n=90	0.29× n=90
	cheap_fno	0.74× n=192	1.38× n=192	0.23× n=192	0.74× n=252	1.38× n=252	0.21× n=252	0.74× n=32	1.38× n=33	0.24× n=30	0.74× n=90	1.38× n=90	0.21× n=90
	deep_fno	1.00× n=192	1.86× n=192	0.31× n=192	1.00× n=252	1.86× n=252	0.29× n=252	1.00× n=33	1.85× n=32	0.32× n=30	1.00× n=90	1.87× n=90	0.29× n=90
source_embedded	direct_fno	0.54× n=192	1.00× n=192	0.17× n=192	0.54× n=252	1.00× n=252	0.15× n=252	0.54× n=32	1.00× n=36	0.17× n=32	0.54× n=90	1.00× n=90	0.15× n=90
	c1_rank0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
	c1_rank1	0.51× n=192	0.94× n=192	0.26× n=288	0.49× n=252	0.92× n=252	0.14× n=288	0.50× n=21	0.93× n=24	0.16× n=30	0.49× n=78	0.92× n=78	0.14× n=78
	c1_rank2	0.30× n=192	0.56× n=192	0.15× n=288	0.29× n=252	0.54× n=252	0.08× n=288	0.30× n=20	0.55× n=22	0.09× n=30	0.29× n=78	0.55× n=78	0.08× n=78
	c2_rank2	0.25× n=192	0.47× n=192	0.13× n=288	0.25× n=252	0.46× n=252	0.07× n=288	0.25× n=20	0.46× n=22	0.08× n=30	0.25× n=78	0.46× n=78	0.07× n=78
	source_embedded	0.24× n=192	0.44× n=192	0.10× n=288	0.23× n=252	0.43× n=252	0.07× n=288	0.23× n=21	0.43× n=24	0.07× n=30	0.23× n=78	0.43× n=78	0.07× n=78
cheap fno deep fno best classical cheap fno deep fno best classical cheap fno deep fno best classical cheap fno deep fno best classical													

Scope and limits

Both RMS and maximum error must be ≤ 2e-05. n is eligible paired measurement cells, not independent fields. Discrete: 24 independent fields; continuum: 12.

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HORIZON-MATCHED COMPARISONS / 2e-06

Offline reanalysis of existing timings: same target track, same final time, same parent / grid / paired seed.

Median comparator latency ÷ candidate latency. Above 1 = candidate faster. Post-hoc feasible schedule selection; excludes offline training cost.

DISCRETE · T=0.27			DISCRETE · T=0.81			CONTINUUM · T=0.27			CONTINUUM · T=0.81				
source_postcompression	source	1.02× n=72	1.90× n=72	0.32× n=72	1.02× n=120	1.91× n=120	0.29× n=120	1.04× n=6	1.92× n=6	0.31× n=6	1.03× n=54	1.91× n=54	0.29× n=54
	rank1	0.55× n=72	1.02× n=72	0.18× n=112	0.55× n=120	1.02× n=120	0.16× n=200	NA	NA	NA	0.55× n=48	1.02× n=48	0.16× n=48
	rank2	0.41× n=72	0.76× n=72	0.13× n=115	0.40× n=120	0.75× n=120	0.12× n=198	NA	NA	NA	0.40× n=48	0.75× n=48	0.11× n=48
	source_trust	0.92× n=72	1.70× n=72	0.29× n=72	0.92× n=120	1.71× n=120	0.26× n=123	0.97× n=6	1.75× n=6	0.28× n=6	0.92× n=42	1.71× n=42	0.26× n=42
	source_consistency	0.80× n=72	1.49× n=72	0.25× n=72	0.80× n=120	1.50× n=120	0.23× n=120	NA	NA	NA	0.80× n=54	1.50× n=54	0.23× n=54
	source_df_loss	1.02× n=72	1.90× n=72	0.32× n=72	1.03× n=120	1.91× n=120	0.29× n=120	1.05× n=6	1.91× n=6	0.31× n=6	1.03× n=54	1.92× n=54	0.29× n=54
	source_df_loss_shared	1.02× n=72	1.90× n=72	0.32× n=72	1.03× n=120	1.91× n=120	0.29× n=120	1.04× n=6	1.91× n=6	0.31× n=6	1.02× n=54	1.91× n=54	0.29× n=54
	source_multiband	1.03× n=72	1.90× n=72	0.32× n=72	1.03× n=120	1.91× n=120	0.29× n=120	1.03× n=6	1.90× n=6	0.31× n=6	1.03× n=54	1.91× n=54	0.29× n=54
	cheap_fno	0.74× n=72	1.38× n=72	0.23× n=72	0.74× n=120	1.38× n=120	0.21× n=123	0.78× n=6	1.41× n=6	0.23× n=6	0.74× n=54	1.38× n=54	0.21× n=54
	deep_fno	1.00× n=72	1.85× n=72	0.32× n=72	1.00× n=120	1.86× n=120	0.29× n=120	1.00× n=6	1.83× n=6	0.29× n=6	1.00× n=54	1.86× n=54	0.28× n=54
	best_classical	0.54× n=72	1.00× n=72	0.17× n=72	0.54× n=120	1.00× n=120	0.15× n=120	0.55× n=6	1.00× n=6	0.16× n=6	0.54× n=54	1.00× n=54	0.15× n=54
	direct_fno	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
source_embedded	c1_rank0	0.50× n=72	0.92× n=72	0.26× n=197	0.49× n=120	0.92× n=120	0.53× n=288	NA	NA	0.56× n=6	0.49× n=48	0.92× n=48	0.14× n=48
	c1_rank1	0.30× n=72	0.55× n=72	0.20× n=200	0.29× n=120	0.54× n=120	0.31× n=288	NA	NA	0.33× n=6	0.29× n=48	0.54× n=48	0.08× n=48
	c1_rank2	0.25× n=72	0.46× n=72	0.17× n=197	0.25× n=120	0.46× n=120	0.26× n=288	NA	NA	0.28× n=6	0.25× n=48	0.46× n=48	0.07× n=48
	c2_rank2	0.23× n=72	0.43× n=72	0.16× n=202	0.23× n=120	0.43× n=120	0.25× n=288	NA	NA	0.26× n=6	0.23× n=48	0.43× n=48	0.07× n=48
	source_embedded	0.41× n=72	0.76× n=72	0.16× n=240	0.12× n=120	0.23× n=120	0.13× n=288	NA	NA	NA	0.04× n=48	0.07× n=48	0.01× n=48
cheap fno			deep fno	best classical	cheap fno	deep fno	best classical	cheap fno	deep fno	best classical	cheap fno	deep fno	best classical

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